

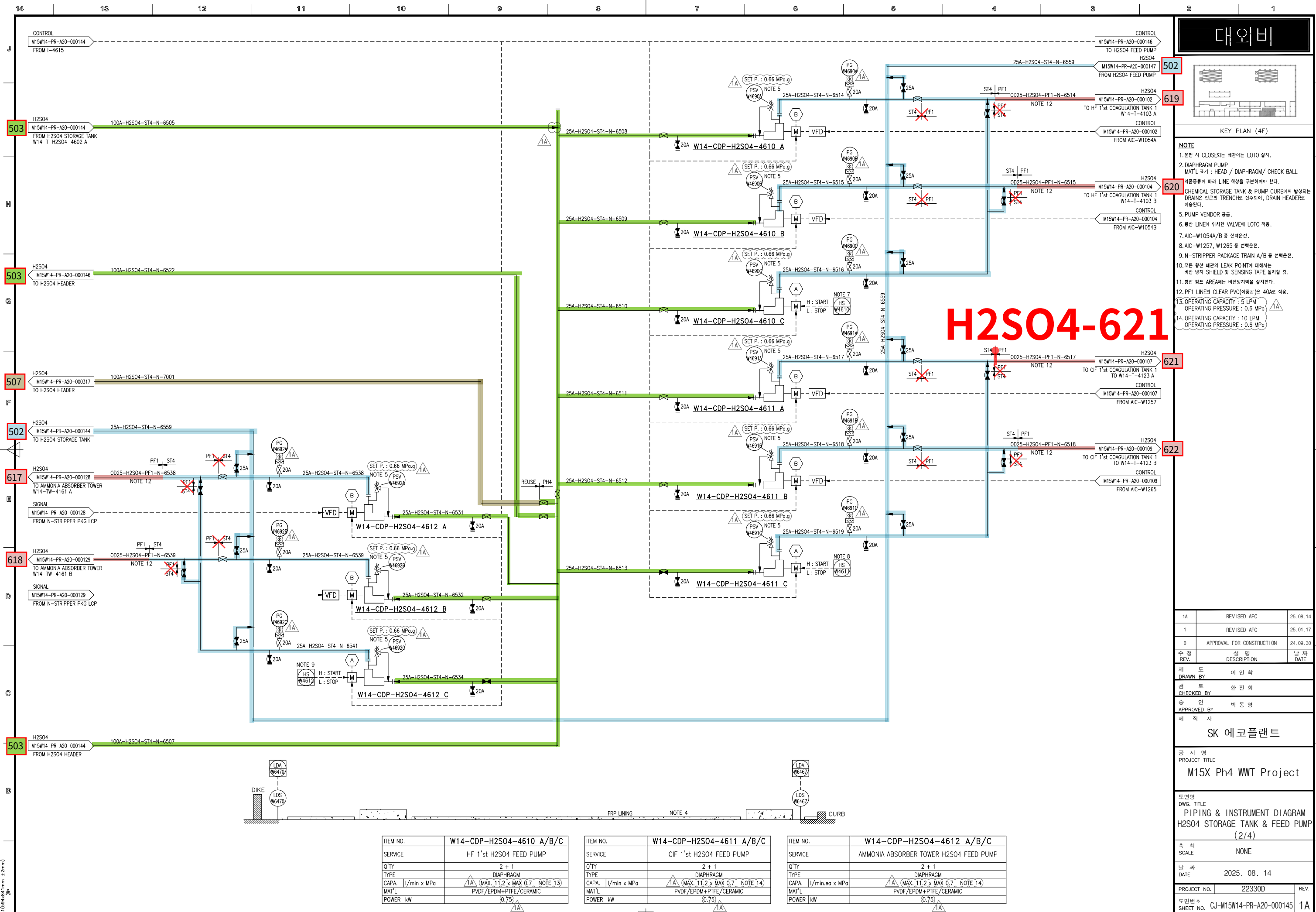
DRAWING LIST FOR TEST PACKAGE

H2SO4-621

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	621	4FA	OD25-H2SO4-PF1-N-6517-01	1 OF 2	0	CJ-M15W14-PR-A20-000107/145	1A/1A	
2	621	4FA	OD25-H2SO4-PF1-N-6517-02	2 OF 2	0	CJ-M15W14-PR-A20-000107/145	1A/1A	
3	621	5FA	OD25-H2SO4-PF1-N-6517-01	1 OF 1	0	CJ-M15W14-PR-A20-000107/145	1A/1A	
4	621	RFA	OD25-H2SO4-PF1-E-6517-01	1 OF 3	1	CJ-M15W14-PR-A20-000107/145	1A/1A	
5	621	RFA	OD25-H2SO4-PF1-E-6517-02	2 OF 3	1	CJ-M15W14-PR-A20-000107/145	1A/1A	
6	621	RFA	OD25-H2SO4-PF1-E-6517-03	3 OF 3	1	CJ-M15W14-PR-A20-000107/145	1A/1A	



대외비

KEY PLAN (4F)

NOTE

1. 온천 시 CLOSE되는 배관에는 LOTO 설치.

2. DIAPHRAGM PUMP MAT'L 표기 : HEAD / DIAPHRAGM/ CHECK BALL

악물종류에 따라 LINE 색상을 구분하여야 한다.

5. PUMP VENDOR 공급.

6. 황산 LINE에 위치한 VALVE에 LOTO 적용.

7. AIC-W1054A/B 중 선택호전.

8. AIC-W1257, W1265 중 선택호전.

9. N-STRIPPER PACKAGE TRAIN A/B 중 선택호전.

10. 모든 황산 배관의 LEAK POINT에 대해서는 비산 방지 SHIELD 및 SENSING TAPE 설치할 것.

11. 황산 탱크 AREA에는 비산방지막을 설치한다.

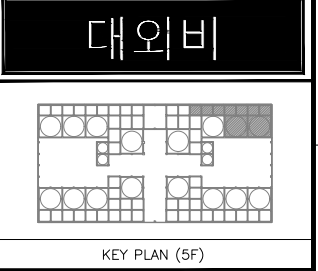
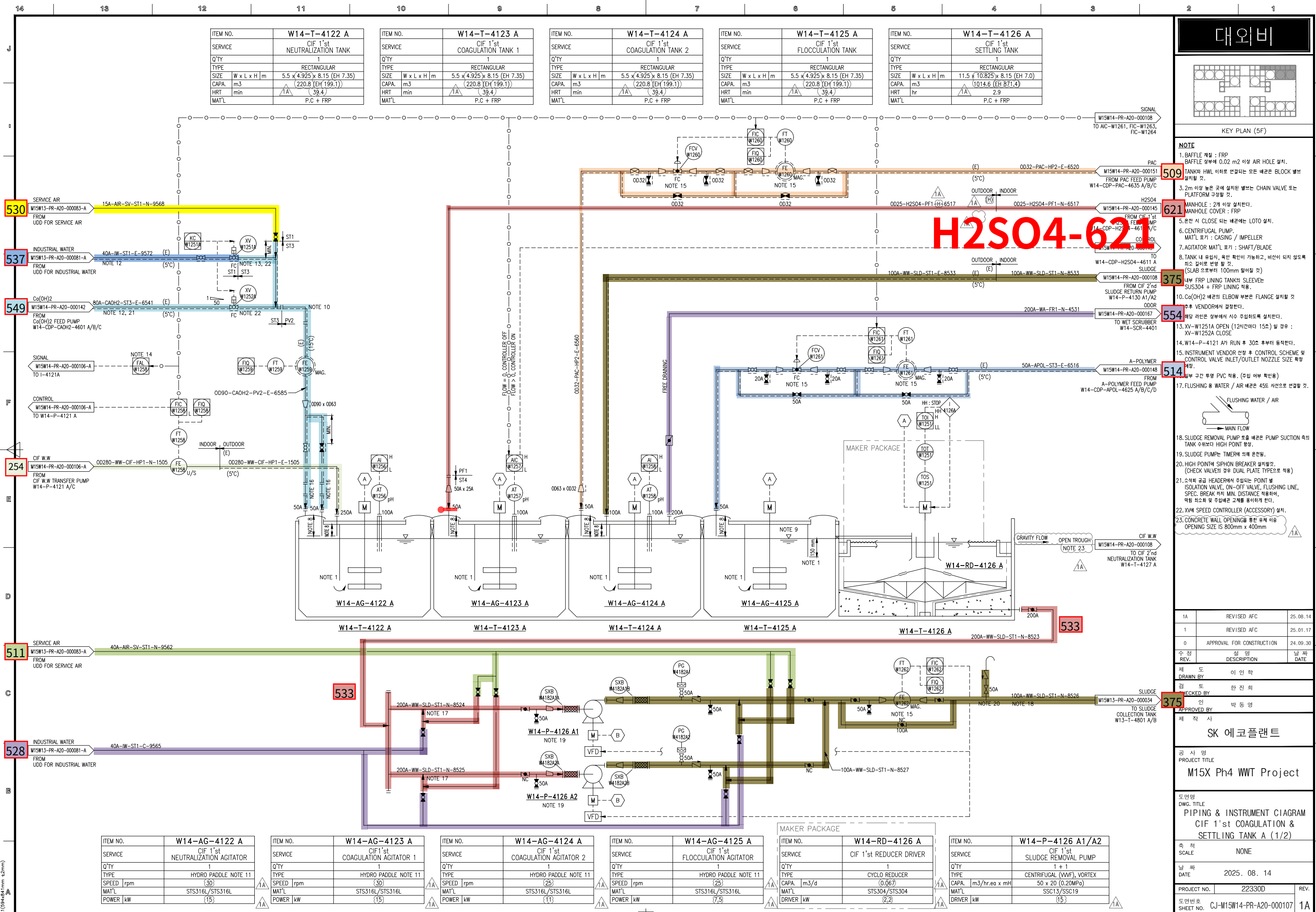
12. PF1 LINE의 CLEAR PVC(이동관)은 40A로 적용.

13. OPERATING CAPACITY : 5 LPM
OPERATING PRESSURE : 0.6 MPa

14. OPERATING CAPACITY : 10 LPM
OPERATING PRESSURE : 0.6 MPa

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정 REV.	설명 DESCRIPTION	날짜 DATE
제 도 DRAWN BY	이 인 학	
검 토 CHECKED BY	한 진 희	
승 인 APPROVED BY	박 동 영	
제 작 사	SK 에코플랜트	
공 사 명 PROJECT TITLE	M15X Ph4 WWT Project	
도면명 DWG. TITLE	PIPING & INSTRUMENT DIAGRAM H2SO4 STORAGE TANK & FEED PUMP (2/4)	
속 척 SCALE	NONE	
날 짜 DATE	2025. 08. 14	
PROJECT NO.	22330D	REV.
도면번호 SHEET NO.	CJ-M15W14-PR-A20-000145	1A

ITEM NO.	W14-CDP-H2SO4-4610 A/B/C	ITEM NO.	W14-CDP-H2SO4-4611 A/B/C	ITEM NO.	W14-CDP-H2SO4-4612 A/B/C
SERVICE	HF 1'st H2SO4 FEED PUMP	SERVICE	CIF 1'st H2SO4 FEED PUMP	SERVICE	AMMONIA ABSORBER TOWER H2SO4 FEED PUMP
Q'TY	2 + 1	Q'TY	2 + 1	Q'TY	2 + 1
TYPE	DIAPHRAGM	TYPE	DIAPHRAGM	TYPE	DIAPHRAGM
CAPA. l/min x MPa	1A (MAX. 11.2 x MAX 0.7 NOTE 13)	CAPA. l/min x MPa	1A (MAX. 11.2 x MAX 0.7 NOTE 14)	CAPA. l/min.ea x MPa	1A (MAX. 11.2 x MAX 0.7 NOTE 14)
MAT'L	PVDF/EPDM+PTFE/CERAMIC	MAT'L	PVDF/EPDM+PTFE/CERAMIC	MAT'L	PVDF/EPDM+PTFE/CERAMIC
POWER kW	(0.75)	POWER kW	(0.75)	POWER kW	(0.75)



- NOTE**
1. BAFFLE 제형 : FRP
BAFFLE 상부에 0.02 m2 이상 AIR HOLE 설치.
TANK의 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것.
 2. 3.2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
 3. MANHOLE : 2개 이상 설치한다.
MANHOLE COVER : FRP
 4. 5. 온전 시 CLOSE 되는 배관에는 LOTO 설치.
 5. 6. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
 6. 7. AGITATOR MAT'L 표기 : SHAFT/BLADE
 7. 8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록 최소 길이로 변경 할 것.
(SLAB으로부터 100mm 떨어질 것)
 8. 9. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.
 9. 10. Co(OH)2 배관의 ELBOW 부분은 FLANGE 설치할 것
 10. 주후 VENDOR에서 결정한다.
 11. 해당 라인은 상부에서 시수 주입하도록 설치한다.
 12. 13. XV-W1251A OPEN (12시간마다 15초) 일 경우 :
XV-W1252A CLOSE
 13. 14. W14-P-4121 A가 RUN 후 30초 후부터 동작한다.
 14. 15. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정
 15. 16. 일부 구간 투명 PVC 적용. (주입 여부 확인용)
 16. 17. FLUSHING 용 WATER / AIR 배관은 45도 서선으로 연결할 것.
 17. 18. SLUDGE REMOVAL PUMP를 배관은 PUMP SUCTION 속의 TANK 수위보다 HIGH POINT 형성.
 18. 19. SLUDGE PUMP는 TIMER에 의해 운전됨.
 19. 20. HIGH POINT에 SIPHON BREAKER 설치할것.
(CHECK VALVE의 경우 DUAL PLATE TYPE로 적용)
 20. 21. 소액의 공금 HEADER에서 주입되는 POINT 별 ISOLATION VALVE, ON-OFF VALVE, FLUSHING LINE, SPEC. BREAK 까지 MIN. DISTANCE 적용하여, 역침 최소화 및 주입배관 고체를 용이하게 한다.
 21. 22. XVM SPEED CONTROLLER (ACCESSORY) 설치.
 22. 23. CONCRETE WALL OPENING을 통한 유체 이용 OPENING SIZE IS 800mm x 400mm

1A	REVISED AFC	25.08.14
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0	APPROVAL FOR CONSTRUCTION	24.09.30
수정	실명	날짜
REV.	DESCRIPTION	DATE

제 도	인 인 학
DRAWN BY	
검 토	한 진 회
CHECKED BY	
인	박 동 영
APPROVED BY	
제 작 사	

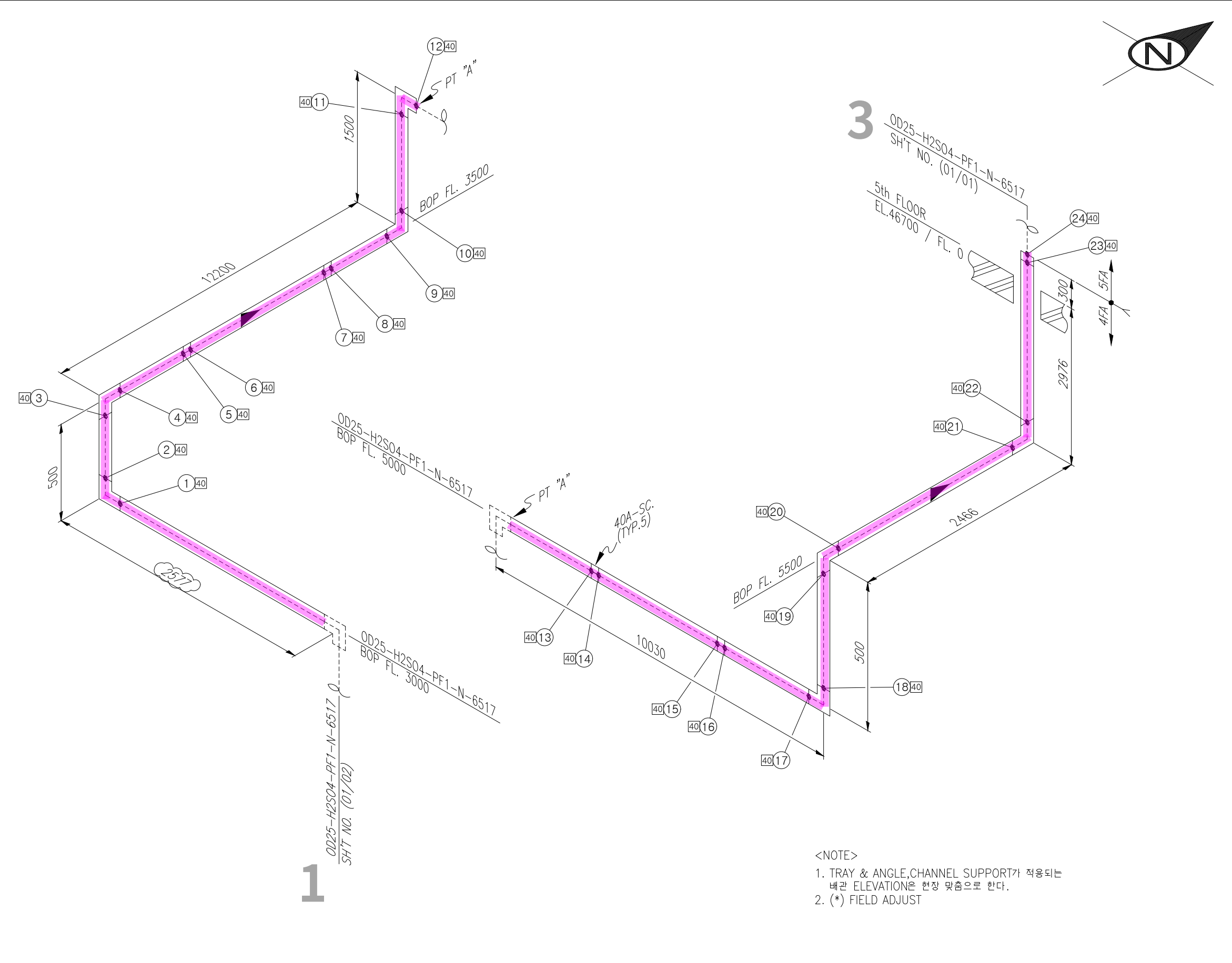
SK 에코플랜트		
공 사 명		
PROJECT TITLE		
M15X Ph4 WWT Project		
도면명		
DWG. TITLE		
PIPING & INSTRUMENT CIAGRAM		
CIF 1'st COAGULATION & SETTLING TANK A (1/2)		
속 척	NONE	
SCALE		
날 짜	2025. 08. 14	
DATE		
PROJECT NO.	22330D	REV.
도면번호	CJ-M15W14-PR-A20-000107	1A
SHEET NO.		



1

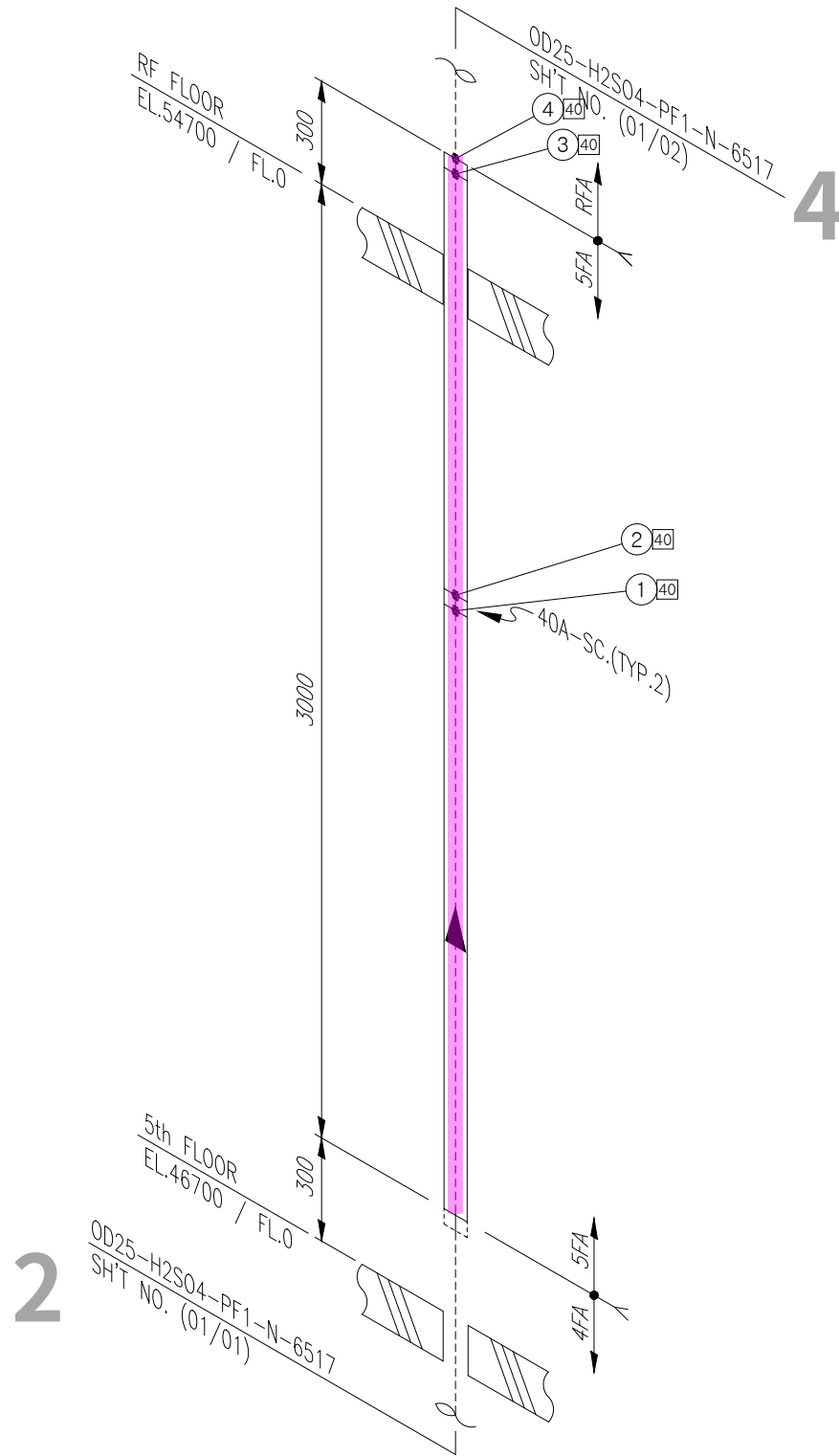
TOR  SK ecoplant

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14									CONTRACTOR	PROJECT NAME
	THICK.	—	DESIGN	6.9	60	UNIT	4FA										M15X Ph4 Project
				OPERATING	6.2	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040008-A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	H2SO4	JOB NO.
						REF. P & ID CJ-M15W14-PR-A20-000145		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	OD25-H2SO4-PF1-N-6517	SH'T NO. 02 / 02	REV. 0

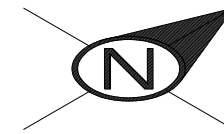
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<NOTE>

1. (*) FIELD ADJUST
2. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	6.7	60	UNIT 5FA							M15X Ph4 Project		
			OPERATING	6.1	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050008-A							COMMODITY	H2SO4	
								25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	0D25-H2SO4-PF1-N-6517	
						REF. P & ID CJ-M15W14-PR-A20-000145	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	SH'T NO. 01 / 01	REV. 0	

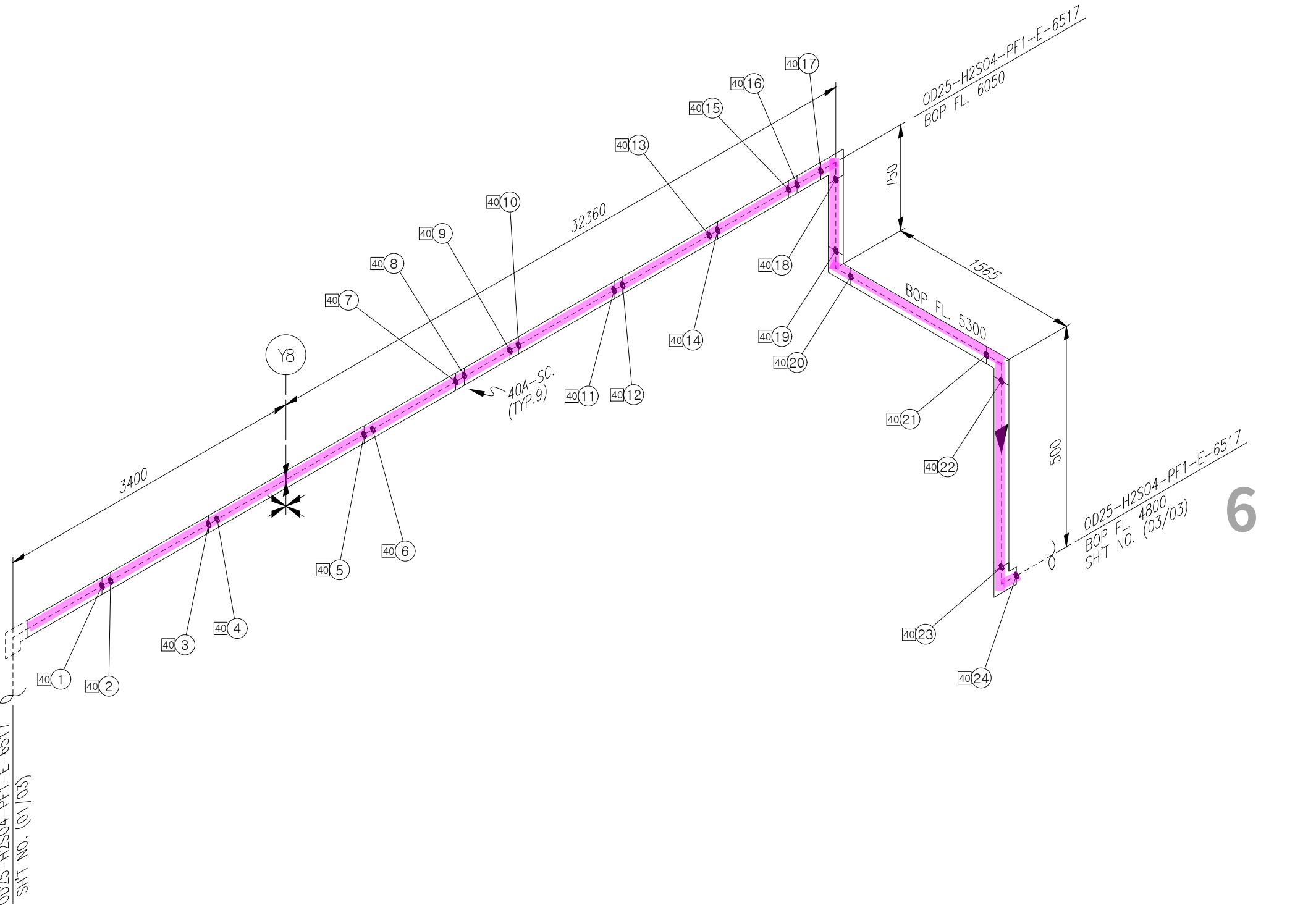


1. (*) FIELD ADJUST



INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	25	DESIGN	6.9	60	UNIT RFA						
			OPERATING	6.2	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10008-A		25.03.31	REVISED AS MARKED	황래중	안남 후	박동
						REF. P & ID		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남 후	박동
						CJ-M15W14-PR-A20-000107	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP

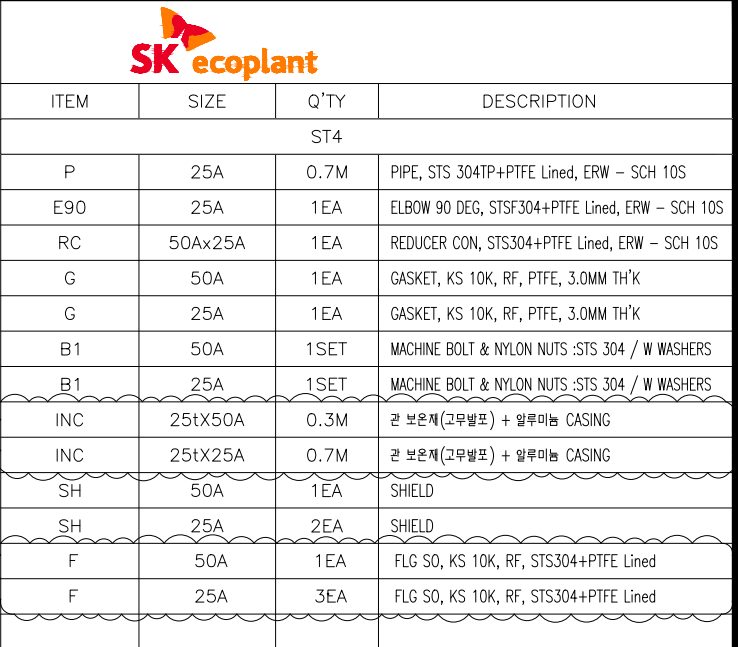
경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

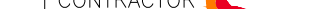
INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14									CONTRACTOR	PROJECT NAME
	THICK.	25	DESIGN	6.9	60	UNIT	RFA										
			OPERATING	6.2	AMB	REF. PLAN DWG.		1	25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	H2SO4	JOB NO.	22330D
						CJ-M15W14-PI-A30-R10008-A		0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영				
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	OD25-H2SO4-PF1-E-6517	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000107											



* EMERGENCY SHOWER : SSH

* SPECIAL BOLT & NUT : BL/BL1

6

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	25	DESIGN	6.9	60	UNIT	RFA								M15X Ph4 Project	
			OPERATING	6.2	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10008-A			25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	H2SO4	JOB NO.
						REF. P & ID CJ-M15W14-PR-A20-000107			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	0D25-H2SO4-PF1-E-6517	SH'T NO. 03 / 03
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			REV. 1